A COMPLETE INTRODUCTION

UNDERSTANDING POLARITY LAB

From zero knowledge to full understanding.

This deck explains everything: the science, the products, the patents, the math.

Polarity Lab is a research laboratory that builds products.

THINK OF IT LIKE THIS:
Imagine a university research lab that also makes apps. The apps generate data. The research studies that data. The research improves the apps. Repeat.

We're studying **how conversation reveals how your brain organizes information**. And we're building products that use this research.

THE RESEARCH SIDE

Conversational Connectomics (CCX)

A computational framework for extracting cognitive signals from natural language. Two patents. Twelve experiments. One 258-page textbook.

THE PRODUCT SIDE

Four Apps, One Platform

Waxfeed (music), Polarity (memory), AVDP (podcast), Pain Points (operations). Each generates data. Each tests our research.

Legal structure: Polarity Lab LLC (Delaware). Research lab with commercial subsidiaries. Theodore Addo is sole owner (100%). Nathan and Shadrack participate in project profits.

Can we measure how your brain organizes knowledge - just from how you talk?

ANALOGY:
A doctor can learn about your lungs by listening to you breathe (with a stethoscope).

We're asking: Can we learn about your *brain's knowledge structure* by listening to you *talk* (with algorithms)?

If the answer is yes, conversation becomes a new source of data about cognition - non-invasive, continuous, longitudinal.

If the answer is no, we'll have learned something important about the limits of language-based cognitive assessment. Either way, valuable science.

Current cognitive assessment is limited.

To check if someone's memory is declining, doctors use tests like the MMSE (Mini-Mental State Exam) or MoCA (Montreal Cognitive Assessment).

These tests are:

- Done once a year (at most)
- In clinical settings only
- Snapshot, not longitudinal
- Can miss subtle early changes

THE PROBLEM:
By the time cognitive decline is detectable on standard tests, significant damage may have already occurred. Early detection matters.

THE OPPORTUNITY

What if we could track cognitive signals *between* clinical visits?

What if subtle changes in how someone talks - the connections they make, the patterns they use - could signal early issues?

THE HYPOTHESIS

Conversational patterns reflect underlying cognitive organization. Changes in cognitive organization show up in conversational patterns. If we can measure the patterns, we might detect the changes.

STATUS CHECK

This is a **hypothesis**, not a proven claim. That's why we're doing the research.

Conversational Connectomics (CCX) is our computational framework.

BREAKING DOWN THE NAME:
Connectomics = the study of neural connections (like mapping wires in the brain)
Conversational = we're doing this from conversation, not brain scans

So "Conversational Connectomics" = inferring how concepts are connected in someone's mind, using their natural language.

WHAT CCX DOES

1.	**Parses**	Breaks conversation into atomic cognitive events
2.	**Classifies**	Labels what type of thinking each event represents
3.	**Graphs**	Builds a network of connected concepts
4.	**Weights**	Calculates how strongly concepts are linked
5.	**Fingerprints**	Generates a unique signature for how you think

THE KEY CONSTRUCTS

UNITS OF COGNITION (UoC)
12 types of cognitive events we detect in speech

MEG-PROXY STATES
12 cognitive "modes" you shift between in conversation

POLARITY POINTS (PP)
A number (0-100) showing how strongly two concepts connect

BRAINID
Your unique cognitive "fingerprint" from how you think

Each of these is explained in detail on the following slides.

Every sentence you say represents a type of cognitive event.

When you speak, you're not just producing words - you're performing cognitive operations. We classify these into 12 types.

ANALOGY: Just like a doctor classifies chest sounds (wheeze, crackle, murmur), we classify utterances (assertion, question, emotion, plan...)

ASSERTION

"I'm from Providence."

Stating a fact. Declarative knowledge.

ACTION

"I'm working on the app."

Describing current activity.

QUESTION

"What time is the meeting?"

Seeking information.

PLAN

"Tomorrow I'll finish the slides."

Future intention, prospective memory.

EMOTION

"I'm so stressed about this."

Expressing affective state.

EVALUATION

"That design is much better."

Judgment, opinion, preference.

REQUEST

"Can you help me with this?"

Seeking assistance.

MEMORY_REF

"Remember when we did X?"

Referencing past experience.

REPAIR

"Wait, I meant Tuesday."

Self-correction. Error monitoring.

SOCIAL

"Thanks so much!"

Social bonding, greetings.

META

"Earlier you said..."

About the conversation itself.

INTENTION

"I want to learn Spanish."

Goals, desires, motivations.

Scientific basis: Speech act theory (Austin 1962, Searle 1969) + dialogue act taxonomies (Stolcke et al. 2000). Not invented from scratch - built

Your brain shifts between cognitive "modes" during conversation.

WHAT IS MEG?
MEG (Magnetoencephalography) is a brain scanning technique that measures magnetic fields from neural activity. Researchers use MEG + statistical methods (HMMs) to identify recurring "brain states" - patterns that come and go.

WHAT IS A PROXY?
We can't do MEG during normal conversation. But we hypothesize that conversational patterns *reflect* those brain states. So we call our language-based detection "MEG-Proxy" - it's a stand-in for what MEG would show.

GROUNDING

"I'm at the coffee shop working."

Setting context. Where/what/when.

NARRATIVE_RECALL

"Last summer we went to..."

Telling stories. Episodic memory.

PLANNING

"Next week I'm going to..."

Future thinking. Prospective.

DECISION

"Should I take the job?"

Weighing options. Choice.

HELP_SEEKING

"Can you explain this?"

Requesting assistance.

SOCIAL_BONDING

"I really appreciate you."

Building connection.

REPAIR

"Wait, let me clarify..."

Correcting. Error monitoring.

META

"This conversation is..."

About the discourse itself.

EVALUATION

"That was really good."

Making judgments.

EMOTIONAL

"I'm feeling overwhelmed."

Processing affect.

RETRIEVAL

"What was that called?"

Memory search.

CREATIVE

"What if we tried..."

Brainstorming. Imagination.

A number that represents how strongly two concepts are connected in your mind.

ANALOGY:
Think of your brain as a city. Concepts are neighborhoods. Some neighborhoods are connected by highways (strong links). Others by side streets (weak links). **Polarity Points (PP)** measures the "highway strength" between any two concepts - based on how often you mention them together and in what contexts.

THE FORMULA (SIMPLIFIED)

```
PP = f(R, C, T, Q)

Where:
R = Reinforcement count (how often mentioned together)
C = Context specificity (do they appear in specific contexts?)
T = Recency (when was the last mention?)
Q = Quality of evidence (how clear was the connection?)
```

The formula uses Bayesian statistics (Beta-Bernoulli estimation) to give us uncertainty bounds, not just a point estimate.

WHAT THE NUMBERS MEAN

PP 90-100	Very strong link (e.g., "mom" ↔ "love")
PP 70-90	Strong link (e.g., "work" ↔ "stress")
PP 50-70	Moderate link (frequent co-occurrence)
PP 30-50	Weak link (occasional mention)
PP 10-30	Very weak (mentioned once or twice)
PP 0-10	Negligible (noise)

Example: If you mention "coffee" and "morning" together 15 times, always in "grounding" state, PP might be ~75. If you mention "coffee" and "interview" once, PP might be ~12.

Why "Polarity"? The name refers to the *direction* of the link (A→B vs B→A) and the *valence* (positive/negative emotional tone). We track both.

A unique cognitive fingerprint generated from how you think.

ANALOGY:
Your fingerprint is unique because of the specific pattern of ridges. Your **BrainID** is unique because of the specific pattern of how you transition between cognitive states and connect concepts. No two people think exactly the same way.

WHAT GOES INTO A BRAINID

1.	**State transitions**	Your pattern of moving between MEG-Proxy states
2.	**PP distribution**	The shape of your knowledge graph (hub nodes, clusters)
3.	**Temporal patterns**	When you think about what (time-of-day, context)
4.	**UoC preferences**	Do you ask more questions? Make more plans?

THE KEY INSIGHT

BrainID should be **stable over time** for healthy individuals. If it changes significantly, that might indicate cognitive change.

APPLICATIONS

AUTHENTICATION (Patent #1)

Your BrainID as a "password" - no one else thinks like you. And it can tolerate natural drift (you don't think exactly the same today as 5 years ago).

LONGITUDINAL TRACKING

If your BrainID shifts suddenly, that's a signal. Could be stress, could be something else. Worth investigating.

COMPATIBILITY

Compare two BrainIDs to see if people "think alike." AVDP uses this to match podcast guests.

Format: BrainID is stored as a vector of numbers (the fingerprint). For users, we display it as: `POL-A7K2-M9P4-Q3R8`

Rules for which memories get promoted to long-term storage.

NEUROSCIENCE BACKGROUND:
In real brains, not every experience becomes a long-term memory. The hippocampus acts as a "gate" - only important, reinforced, emotionally significant memories get consolidated. We mimic this process computationally.

THE STAGES

LABILE	New memory. Easily forgotten. First 24 hours.
STABILIZING	Getting stronger. 1-7 days. Needs reinforcement.
CONSOLIDATED	Stable. 7-30 days. Resistant to interference.
PERMANENT	Long-term. 30+ days. High PP. Integrated.
FADING	Not reinforced. Starting to decay.
ARCHIVED	Very low PP. Stored but not easily accessible.

THE RULES

To move from LABILE → STABILIZING:
PP must be > 15 AND memory must be > 1 day old

To move from STABILIZING → CONSOLIDATED:
PP must be > 40 AND memory must be > 7 days old AND reinforced 3+ times

To move from CONSOLIDATED → PERMANENT:
PP must be > 70 AND memory must be > 30 days old AND reinforced 10+ times

DECAY: Memories not reinforced for 14 days start FADING. After 30 days without reinforcement, they get ARCHIVED.

Why this matters: The Polarity app surfaces memories that need reinforcement before they fade. It's like spaced repetition, but for your personal knowledge graph.

APPLICATION #63/940,728 • FILED DEC 2025 • 11 CLAIMS

System and Method for Neuroanatomically-Grounded Knowledge Graph Construction and Cognitive Biometric Authentication

WHAT IT COVERS

KNOWLEDGE GRAPH CONSTRUCTION

Building a graph of concepts from conversation, where edges are weighted by how the brain would connect them (based on neuroscience literature).

COGNITIVE FINGERPRINT

Generating a unique identifier (BrainID) from the structure and dynamics of the knowledge graph.

N-BKDF AUTHENTICATION

Using the knowledge graph itself as a "password." Questions about your own life that only you can answer correctly.

DRIFT-TOLERANT AUTH

Authentication that tolerates the fact that your knowledge evolves. You don't think exactly the same as 5 years ago - the system adapts.

SHADRACK'S CONTRIBUTION

Shadrack originated the key insight: **what if the knowledge graph IS the password?**

Traditional passwords are static. You memorize them. But your knowledge is dynamic - it grows and evolves. Shadrack proposed using this as an "evolvable cryptography scheme."

THIS LED TO:

- N-BKDF (Neural Biometric Key Derivation Function)
- Drift-Tolerant Authentication
- BrainID generation

Example: Instead of "What's your password?", the system asks "What did you and your mom talk about last Thanksgiving?" - a question only you can answer, drawn from your own knowledge graph.

APPLICATION #63/960,633 • FILED JAN 2026 • 12 CLAIMS

Conversational Connectomics: Methods for Bayesian Associative Strength Estimation, MEG-Proxy Cognitive State Classification, and Integrated Cognitive Fingerprinting from Natural Language Dialogue

WHAT IT COVERS

POLARITY POINTS (CLAIMS 1-4)

The mathematical formula for calculating edge strength. Bayesian Beta-Bernoulli estimation. Context specificity. Recency decay.

UNIT OF COGNITION (CLAIMS 5-6)

Segmenting conversation into atomic cognitive events. The 12-type classification system.

MEG-PROXY STATES (CLAIMS 7-8)

The 12-state cognitive mode model. State transition tracking. HMM-inspired inference.

CONSOLIDATION GATE (CLAIMS 9-10)

The memory promotion rules. Stage transitions. Decay functions.

COGNITIVE FINGERPRINT (CLAIMS 11-12)

Generating BrainID from state transitions + graph topology. Proactive memory surfacing.

NATHAN'S CONTRIBUTION

Nathan helped formalize CCX as a **research field**, not just a product feature.

His work on the claim structure - distinguishing computational methods from theoretical framework - made the patent defensible.

WHY THIS MATTERS

Patent #1 covers the *application* (Polarity product). Patent #2 covers the *science* (CCX framework). Together, they protect both the technology and the methodology.

> **Protection:** If someone builds a different app using similar methods (Bayesian edge weighting, HMM state classification for cognition), Patent #2 applies. If they copy our specific product, Patent #1 applies.

You can't claim "this detects Alzheimer's" without evidence. Each level requires more proof.

WHY A LADDER?
Science doesn't jump from "idea" to "proven." You climb through levels of evidence. We're explicit about where we are on that climb.

THE SIX LEVELS

L0	**Speculation**	Just an idea. No testing.
L1	**Theoretical**	Makes logical sense. Supported by literature.
L2	**Computational**	**← WE ARE HERE**
L3	**Behavioral**	Correlates with clinical tests (MMSE, MoCA)
L4	**Neural**	Brain imaging confirms (fMRI, MEG)
L5	**Causal**	Proven mechanism. Intervention studies.

L2: WHERE WE ARE

WHAT L2 MEANS

The algorithms work. We can classify UoC. We can detect states. We can calculate PP. We can generate BrainID.

Current accuracy:

- UoC classification: 69% (baseline 8%)
- State classification: 58% (baseline 8%)

What this DOESN'T mean: Clinical utility. Diagnostic validity. Medical claims. Those require L3+.

NEXT: L3 BEHAVIORAL VALIDATION

Run studies where we compare CCX metrics to established clinical assessments. If Polarity Points correlates with MMSE scores, we have behavioral validation.

A "cognitive memory service" - a friend who never forgets.

SIMPLE EXPLANATION:
You talk to Polarity like you'd talk to a friend. It remembers what you tell it. But unlike a friend, it organizes those memories, finds connections you might miss, and helps you retrieve things when you need them.

KEY FEATURES

MEMORY EXTRACTION

You share something ("My mom's birthday is March 15"). Polarity extracts the memory, classifies it (ASSERTION), links it to related concepts (mom, family, March).

KNOWLEDGE GRAPH

All your memories become nodes. Connections become edges. Polarity Points weight the edges. You can see how your knowledge is structured.

PROACTIVE SURFACING

Polarity notices when a memory is relevant and brings it up. "You mentioned you wanted to call your sister this week."

CONSOLIDATION

Memories at risk of fading are surfaced for reinforcement. Like spaced repetition for your life.

CCX CONNECTION

The Polarity app is the **primary research platform** for CCX. Every conversation generates data.

Feature	CCX Construct
Memory extraction	UoC classification
Knowledge graph	PP edge weighting
State tracking	MEG-Proxy detection
Proactive surfacing	Predictive surfacing
Memory aging	Consolidation Gate
User fingerprint	BrainID

RESEARCH VALUE

Real users, real conversations, real data. We can measure if the algorithms work in practice.

A social platform for music enthusiasts. Built by Shadrack.

SIMPLE EXPLANATION:
Waxfeed is where music lovers discover, share, and connect around music. DJs post mixes. Fans follow taste profiles. The platform learns what you like and recommends accordingly.

KEY FEATURES

DJ DISCOVERY

Find DJs whose taste aligns with yours. Not by genre - by actual listening patterns.

TASTE GRAPH

Your musical preferences form a graph. Artists, tracks, genres - all connected by how often you engage with them.

SOCIAL LAYERS

See what friends are listening to. Discover through trusted sources, not algorithms alone.

ENTERPRISE PARTNERSHIPS

Platform licensing for radio stations and venues. DJs get paid when their mixes get played.

CCX CONNECTION

Music taste IS a form of cognition. How you organize your preferences reveals cognitive patterns.

Waxfeed Feature	CCX Parallel
Taste graph	Knowledge graph (PP edges)
Listening patterns	Behavioral fingerprint
DJ matching	Compatibility prediction
Recommendation	Predictive surfacing

RESEARCH VALUE

Music preferences are stable, measurable, and emotional. Waxfeed generates CCX-analyzable data in a domain where ground truth is easier to verify.

Principal: Shadrack (50%) • **Partners:** Theodore (25%), Nathan (25%)

A Very Distant Perspective. A podcast that uses CCX.

SIMPLE EXPLANATION:
Most podcasts have one-off guests. AVDP does **deep dives** - multiple consecutive episodes with the same person to learn their cognitive profile. Then we **match** them with other guests based on predicted compatibility.

THE FORMAT

DEEP DIVE SERIES

3-5 episodes with one guest. Different topics each time. The goal: build a comprehensive cognitive profile through natural conversation.

PROFILE BUILDING

We apply CCX to the conversation. Extract their UoC patterns, state transitions, PP graph. Build their BrainID.

MATCHED EPISODES

Based on profiles, we predict who would have a great conversation. Bring them together. Test if the prediction worked.

CCX CONNECTION

AVDP is **CCX in action**. It demonstrates the research publicly.

AVDP Feature	CCX Construct
Deep dive episodes	Data collection
Profile building	BrainID generation
Match prediction	Compatibility algorithm
Match success	Validation data

RESEARCH VALUE

We can measure if CCX predicts conversational compatibility. If matched guests have better conversations than random pairings, CCX has predictive validity.

Principal: Theodore (50%) • **Partners:** Shadrack (25%), Nathan (25%)

The operating philosophy of the entire lab.

THE IDEA:
Traditional labs have one PI, employees, and projects assigned top-down. Pain Points is different: **anyone can originate a project**. If you build the MVP, you become the "Principal" - the leader who gets the highest share. Everyone else who joins becomes a "Partner."

THE FIVE PRINCIPLES

1.	**Lab Fee, Not Extraction**	15% of revenue to Lab Bank (service fee)
2.	**Tiered Reserves**	Build 3mo → 6mo buffer, then 10% forever fund
3.	**Principal Primacy**	Builder gets most. MVP required.
4.	**Domain Lead Autonomy**	Run your project like your company
5.	**Everyone UP**	All banks grow. Every month. No exceptions.

THE LOGIC

If you build something valuable, you deserve the most reward. If you help someone else build, you deserve fair reward. The Lab takes a service fee for infrastructure.

ALLOCATION EXAMPLE: WAXFEED

Revenue: $100,000
Step 1: Lab Fee (15%) = $15,000 → Lab Bank
Step 2: Remaining = $85,000
Step 3: If 3-month reserve not met, retain 100%
Step 4: If reserve met, distribute:
- Shadrack (Principal): 50% = $42,500
- Theodore (Partner): 25% = $21,250
- Nathan (Partner): 25% = $21,250

TITLE HIERARCHY

Associate → Partner → Principal → **Core**

Core status: 3+ successful project originations. Gets lab-level profit share.

FOUNDER & PI

Theodore Addo

MD Candidate, Brown Alpert Medical School '26

10+ years on this question. MIT Boyden Lab alum (expansion microscopy, whole-brain imaging). Radiology-bound.

Role: 100% owner. PI on all research. Domain Lead for Polarity, AVDP, Pain Points.

COFOUNDER

Shadrack Annor

Brown University

Music-tech expertise. Built Waxfeed from scratch. Originated the evolvable cryptography insight (Patent #1).

Role: Principal of Waxfeed (50%). Partner on other projects (25% each). Co-inventor on both patents.

COFOUNDER

Nathan Amankwah

University of Ottawa

Frontend/UX lead. Made Polarity beautiful and usable. Helped formalize CCX as a research field (Patent #2).

Role: Partner on all projects (25% each). Dev Lead. Co-inventor on Patent #2.

OWNERSHIP CLARIFICATION

Theodore owns 100% of Polarity Lab LLC. Nathan and Shadrack participate in project profits through Profit Participation Agreements - not company ownership. They are **cofounders** (startup sense) but not **members** (legal sense). This structure protects the research integrity (PI control) while rewarding contribution fairly.

Four arms. One mission.

RESEARCH LAB

CCX Science

PI structure. Papers. Grants. Patents. The core intellectual property.

STARTUP

First Movers

Products that commercialize the research. Waxfeed, Polarity, AVDP.

VENTURE

Investment Arm

Fund aligned projects. Core status = portfolio company credit.

STUDIO

Communication

AVDP. Content. Scientific storytelling. Public engagement.

THE FLYWHEEL

Products generate **data** →
Data enables **research** →
Research produces **papers** + **patents** →
Papers bring **credibility** →
Credibility brings **users** + **funding** →
Funding enables more **products** →
Repeat.

LEGAL STRUCTURE

Entity: Polarity Lab LLC (Delaware)

Location: Providence, RI

Website: polarity-lab.com

Formed: December 4, 2025

From L2 computational to L4 neural validation.

2026: L3 BEHAVIORAL

Correlate with clinical assessments

- Run study with MMSE/MoCA
- Compare PP scores to memory scores
- If correlation exists, L3 achieved
- If not, understand why

Collaborator needed: Clinical psychologist (L3 validation studies)

2027: L4 NEURAL

Neuroimaging confirmation

- Partner with imaging lab
- Collect conversation + fMRI/MEG
- Does our "MEG-Proxy" match real MEG?
- Yeo 7-Network validation

Collaborator needed: Neuroimaging researcher

2028+: CLINICAL PATH

If validated, pursue clinical utility

- Early cognitive decline detection
- Longitudinal monitoring
- FDA pathway if warranted
- Or: publish null results

Outcome-agnostic: valuable science either way

We're seeking collaborators for L3 and L4 validation. Standard academic collaboration - co-authorship, your methods, our framework.

Not published yet. In progress.

WHAT EXISTS NOW

- 258-page CCX framework document (internal)
- 12 computational experiments run
- 2 provisional patents filed
- Working product with real users

We're at the "write it up" stage, not the "published in Nature" stage.

PAPER PIPELINE

1.	**CCX Framework Paper**	Drafting now
2.	BrainID Validation	Q2 2026
3.	Polarity Points Reliability	Q3 2026
4.	Dyadic CCX Framework	2027
5.	Clinical Preliminary	Post-L3

Target venues: CHI, EMNLP, CogSci, Frontiers

If null results: We publish those too. Science is about truth, not just positive results.

We know this could be misused.

Cognitive monitoring could enable surveillance, coercion, abuse - especially in intimate partner violence scenarios.

WE ARE NOT NAIVE.
Published research on tech-facilitated abuse (e.g., CHI 2018) shows how "safety features" become surveillance tools. We need adversarial review BEFORE deployment.

WHY WE NEED SAFETY COLLABORATORS

We want researchers to actively try to break our safety model, find abuse vectors, and design detection mechanisms. We're seeking collaborators with expertise in tech safety and intimate partner violence research.

SAFETY PRINCIPLES (NON-NEGOTIABLE)

1. **User-only access**	No one sees your data but you
2. **No default sharing**	Caregiver access = explicit consent
3. **Coercion detection**	Flag patterns suggesting forced access
4. **Right to deletion**	Full data purge on request
5. **No predictive policing**	Never used for risk scoring

IRB REQUIREMENT

Any clinical validation requires IRB approval. Product users consent to aggregate analytics. Research requires separate consent.

Multiple ways to participate.

RESEARCH COLLABORATORS

Domain experts for L3/L4 validation. Co-authorship. Your methods, our framework.

ADVISORS

Occasional feedback on direction, ethics, clinical pathway.

PILOT PARTNERS

Institutions deploying CCX tools (with proper IRB/consent).

CONTRIBUTORS (PAIN POINTS MODEL)

Build an MVP for a project you care about. Become Principal.

The immediate ask.

We'd like 30 minutes to discuss whether there's alignment.

WHAT WE'D COVER

1. Your questions about the framework
2. Whether our research interests overlap
3. If there's a collaboration that makes sense

CONTACT

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MEASURE THE MIND

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